CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Experiment 3: Fractional Knapsack Problem using Greedy Strategy

Course: Design Analysis and Algorithms Laboratory [23CS3505]
Category: Experiments / Practicals | Experiment #3 | Academic Year: 2026-27

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1. Overview & Objective:

Maximize total value under capacity constraint by calculating value-to-weight ratios.

2. Practical Tasks & Verification Steps:

1. Define Item class containing value, weight, and calculated value/weight ratio.
2. Sort items in descending order of value/weight ratio using custom comparator.
3. Iterate through sorted items, taking full items while capacity allows and fractional fraction of final item.
4. Output maximum achievable total value and fraction array of taken items.

Submission Guidelines & Verification:

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